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| **Fiscal Year:** | FY 2026 | **Task Last Updated:** | FY 03/13/2026 |
| **PI Name:** | Pereira, Talmo Ph.D. | | |
| **Project Title:** | Automated Deep Learning for Spaceflight Rodent Behavior Quantification and Health Phenotyping | | |
| **Division Name:** | Human Research | | |
| **Program/Discipline:** | | | |
| **Program/Discipline--Element/Subdiscipline:** | | | |
| **Joint Agency Name:** | | **TechPort:** | No |
| **Human Research Program Elements:** | (1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H) | | |
| **Human Research Program Risks:** | (1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders | | |
| **Space Biology Element:** | None | | |
| **Space Biology Cross-Element Discipline:** | None | | |
| **Space Biology Special Category:** | None | | |
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| **City:** | La Jolla | **State:** | CA |
| **Zip Code:** | 92037-1002 | **Congressional District:** | 50 |
| **Comments:** | | | |
| **Project Type:** | Ground | **Solicitation / Funding Source:** | |
| **Start Date:** | 11/15/2023 | **End Date:** | 11/14/2025 |
| **No. of Post Docs:** | 1 | **No. of PhD Degrees:** | 0 |
| **No. of PhD Candidates:** | 0 | **No. of Master' Degrees:** | 0 |
| **No. of Master's Candidates:** | 1 | **No. of Bachelor's Degrees:** | 1 |
| **No. of Bachelor's Candidates:** | 7 | **Monitoring Center:** | NASA JSC |
| **Contact Monitor:** | Whitmire, Alexandra | **Contact Phone:** | |
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| **Flight Program:** | | | |
| **Flight Assignment:** | NOTE: End date changed to 11/14/2025 per L. Juliette/HFBP (Ed., 12/19/24). | | |
| **Key Personnel Changes/Previous PI:** | Co-Is Frederico Kiffer, PhD and Amelia Eisch, PhD have been added to the project for behavioral segmentation and experimental expertise | | |
| **COI Name (Institution):** | Saunders, Lauren Ph.D. ( NASA Ames Research Center )<br>Scott, Ryan M.A. ( NASA Ames Research Center )<br>Eisch, Amelia ( The Children's Hospital of Philadelphia )<br>Kiffer, Frederico ( The Salk Institute for Biological Studies ) | | |
| **Grant/Contract No.:** | 80NSSC24K0346 | | |
| **Performance Goal No.:** | | | |
| **Performance Goal Text:** | | | |

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| **Task Description:** | Spaceflight isolation and confinement increase the risk of behavioral health impairments and pose major hurdles for crewed deep space missions. To reduce these risks, further research is needed on the effects of spaceflight on psychiatric and behavioral health in order to develop systems for monitoring crew health and performance. This proposal aims to address this gap by applying artificial intelligence (AI)-based technology to enable automated behavioral quantification and health monitoring from spaceflight videos.<br>Powered by state-of-the-art deep learning and computer vision technology, the proposed platform will leverage previous work on automated markerless motion capture (i.e., whole body movement tracking from video) and behavioral phenotyping (i.e., detection of behavior events such as walking or eating). This technology has been used for video-based behavioral analysis in insects, fish, plants, rodents, and humans, including for applications such as health monitoring in animal studies of cancer and neurodegeneration.<br><br>The proposed work will demonstrate the feasibility of using this technology to automate spaceflight behavioral health monitoring by applying it to previously collected videos from the NASA Rodent Research-1 mission. These have been painstakingly manually annotated by human expert observers with frame-by-frame labels of behaviors (e.g., feeding, grooming) to enable quantitative analysis of rodent behavior during spaceflight. This laborious effort was necessary due to the challenging imaging conditions inherent in spaceflight videography – but which are overcome through the use of AI in the proposed work. By developing a platform capable of automating this process, we will establish the foundation for future systems that will be able to monitor behavioral health in research missions – automatically and in real-time, opening the door to interventional studies aimed at maintaining positive behavioral health conditions. Future work may adapt this technology for behavioral monitoring in humans to detect and mitigate the risks of crewed deep space missions. |
| **Rationale for HRP Directed Research:** | |
| **Research Impact/Earth Benefits:** | This work is significant to the solicitation objectives as it directly addresses the Human Research Roadmap "Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders." Our work informs the feasibility rodent health checks and disease onset identification in spaceflight environments, potentially saving hundreds of hours of personnel time in future missions. We demonstrate that in-flight behavioral segmentation and kinematic analysis is possible, and that some behaviors indicative of behavioral health can be inferred from video datasets. We also recommend hardware modifications to improve behavioral segmentation and health monitoring.<br>Furthermore, through demonstrating the feasibility of behavioral segmentation and kinematic analysis of mice in spaceflight, this research lays the groundwork for adapting the technology to enable human health and behavioral performance monitoring in upcoming crewed deep space missions. The proposed work is also valuable to the Institutional Animal Care and Use Committee and the Rodent Research Project as it fills a critical technology gap.<br><br>By leveraging existing flight data from the RR-1 mission, this project validates the applicability of AI-based behavioral analysis tools to other existing and upcoming RR missions. This approach not only benefits future missions but also increases the utility of data collected in past missions, supporting NASA's goals of maximizing the scientific return from spaceflight experiments and advancing our understanding of biological responses to the space environment. Our open source pose estimation and behavioral segmentation models are available for use in other RR missions. |
| **Task Progress:** | Spaceflight poses numerous risks to the behavioral health of flight crews. To reduce these risks, further research is needed on the effects of spaceflight on psychiatric and behavioral health in order to develop systems for monitoring crew health and performance. This work aims to address this gap by applying artificial intelligence (AI)-based technology to enable automated behavioral quantification and health monitoring from videos of freely-behaving mice aboard the International Space Station (ISS).<br>Powered by state-of-the-art deep learning and computer vision technology, our platforms leverage previous work on automated markerless motion capture (i.e., whole body movement tracking from video) and behavioral phenotyping (i.e., detection of behavior events such as walking or eating). This technology has been used for video-based behavioral analysis in insects, fish, plants, rodents, and humans, including for applications such as health monitoring in animal studies of cancer and neurodegeneration.<br><br>Here, we evaluated the video dataset from the Rodent Research-1 (RR-1) mission to the ISS for feasibility of AI-based pose estimation and behavioral segmentation. We describe in detail the challenges associated with computer vision-based approaches to automating behavioral capture, and developed solutions to some of these challenges, including resolution normalization and correction for spherical aberration in the lenses of the AEX-M vivarium system aboard the ISS.<br><br>In this work, we first determined inter-annotator variability in pose detection, given the resolution and visual occlusions are challenging, even to human annotators. We determined that very few keypoints (features on the animals' bodies) are reproducibly inferred across human annotators from the full 3-dimensional views of mice in microgravity from the vantage point of a single camera. Therefore, pose estimation was reduced to 3 reliable keypoints: the snout, neck, and tail-torso interval (TTI). Nine (9) human annotators labelled 15 identical frames sampled across the mission duration, resulting in 144 common labels. We found that distances to the consensus label averaged from 3-5 pixels per keypoint. We then curated a human annotated pose estimation dataset, consisting of 3,249 labels across 2,063 frames sampled throughout the mission. We trained a pose estimation model using our pipeline, SLEAP (Pereira et al., 2022). Localization errors averaged to approximately 20 pixels from ground truth per keypoint. Comparing model inferences to the human inter-annotator variability dataset reveals that model localization errors approached human label distances to the consensus, suggesting our pose estimation model performs at near human labeling accuracy.<br><br>Prior work characterized one of the critical limitations to observing behaviors in the RR-1 dataset by detailing the soiling aggregation on the lens throughout the mission (Ronca et al., 2019). To gain a sense of how lens soiling impacts pose estimation predictive accuracy, we calculated the Object Keypoint Similarity (OKS; Ronchi and Perona, 2017), a measure of inference pixel distances to the ground truth across the mission duration. We found that OKS declined markedly after mission day 13.<br><br>One key limitation of the RR-1 video dataset is that many occlusions -- ranging from an internal grid, to soiling accumulation, to animal overlap from the monocular camera perspective -- prevent individual identity tracking, thereby limiting the range of kinematic analysis from pose estimation. However, kinematic analyses are reliable in video |

samples with a single subject in view. An interesting behavioral spaceflight adaptation in the AEM-X vivarium noted by Ronca et al. is a circling behavior where mice run along four walls about a central axis, effectively centrifuging themselves. The RR-1 dataset contains one video sample of a single mouse engaging in this circling behavior on mission day 10. Because a goal of many RR missions is to understand microgravity effects on biological endpoints, and to further probe mouse behavioral adaptations, we devised a pipeline to estimate the centripetal acceleration mice experience in this vivarium of known dimensions. We find that on day 10 of the mission, a mouse experiences highly oscillating centripetal forces ranging from 0 to 12.5 m/s2, occasionally exceeding 1xG. We note that videos of mice in subsequent mission days appear to show even faster circling, but kinematic analyses are prohibitive due to visual occlusions.

Although our 3-keypoint pose estimation model is reliable for gross locomotor analyses, it is prohibitive for generating a behavioral classifier. Therefore, we turned to DeepEthogram, a pipeline that leverages optic flow in lieu of pose estimation for behavioral segmentation (Bohnslav et al., 2022). The tradeoff with this pipeline is that behavioral segmentation occurs at the frame-level, preventing subject-specific analyses. Human annotators first devised reproducible criteria to distinguish behavioral annotation. To curate a frame-level behavioral dataset, 3 annotators labelled 248,940 frames spanning 66 videos across the mission. This behavioral classifier yielded modest performance metrics under the available visual environment. In absence of individual tracking, and a highly skewed dataset (most sampling is provided early on in the mission, and light cycle samples are rare past day 14) empirical analyses of behavioral adaptations are prohibitive. Cumulative analyses suggest that the most commonly detected behaviors are eating, followed by crawling and circling. Fewer inferences of drifting (mouse movement in absence of contact with the walls), grooming, huddling, and backflips are present, in diminishing order. From the available video sampling, we characterize an increase in crawling from days 3 through 14, increased eating from days 6 through 32, increased drifting from days 7 through 13, and increased circling from days 10 through 32, suggesting these may be temporally dependent behavioral adaptations to spaceflight. We stress that these findings, although insightful, are limited to model performance and sampling availability. Analysis of subsequent RR missions with more representative sampling is needed to build on these findings.

Our work demonstrates for the first time the feasibility of using AI-based pose and behavioral segmentation technology to automate spaceflight behavioral health monitoring by applying it to archived videos from the NASA RR-1 mission. By developing and leveraging platforms capable of automating this process, we have established the foundation for future systems that will be able to monitor behavioral health in research missions, automatically and in real-time, opening the door to interventional studies aimed at maintaining positive behavioral health conditions, and rodent health surveillance. Future work may adapt this technology for behavioral monitoring in humans to detect and mitigate the risks of spaceflight to crews.

| **Bibliography Type:** | Description: (Last Updated: 03/17/2026) |
|---|---|
| **Abstracts for Journals and Proceedings** | Kiffer, FC, Mayo A, Martens M, Mendoza M, Gautam S, Huang J, Bathwal M, Jaikumar S, Mahajan A, Scott R, Sanders L, Eisch A, Pereira T. "Markerless pose estimation and automated behavioral tracking of mice during spaceflight on the International Space Station." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025.<br>Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 |
| **Abstracts for Journals and Proceedings** | Martens M, Kiffer, FC, Mayo A, Huang J, Gautam S, Mendoza M, Bathwal M, Jaikumar S, Mahajan A, Scott R, Pereira T, Eisch A. "Markerless pose estimation and automated behavioral tracking of freely behaving mice during spaceflight on the International Space Station." 2025 Annual Meeting of the Philadelphia Chapter of the Society for Neuroscience, Philadelphia, Pennsylvania, May 30, 2025.<br>Abstracts. 2025 Annual Meeting of the Philadelphia Chapter of the Society for Neuroscience, Philadelphia, Pennsylvania, May 30, 2025. , May-2025 |
| **Articles in Peer-reviewed Journals** | Duporge I, Pereira T, Castiello de Obeso S, Bright Ross JG, Lee SJ, Hindle AG. "The utility of animal models to inform the next generation of human space exploration." npj Microgravity. 2025 Feb 22;11:7. Review. https://doi.org/10.1038/s41526-025-00460-5 ; PMID: 39984492; PMCID: PMC11845785 , Feb-2025 |